

VIKAS RAMASWAMY

Software Engineer | Applied AI, Data Engineering & LLM Systems

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vi6120.github.io/portfolio-website

PROFILE

Software Engineer with a data engineering foundation and hands on applied AI experience. 5+ years building SQL, SAP and Python based data pipelines for finance and platform teams at Capgemini, Mercedes-Benz and Mutares, combined with practical work shipping AI systems through personal projects: retrieval augmented generation, embeddings and vector search, LangGraph based multi agent orchestration, and a reliability and governance layer for AI agents, built with FastAPI backends and React frontends. M.Eng. in Business Intelligence & Data Analytics (1.91). Independent, structured and analytical; available immediately in Germany.

TECHNICAL SKILLS

Applied AI & LLM Systems: Retrieval augmented generation, embeddings and vector search (pgvector), LangGraph multi agent orchestration, prompt and context engineering, human in the loop workflow design

Engineering & Data: Python (pandas, automation, FastAPI), Advanced SQL, PostgreSQL, SAP BW / HANA / SLT, dbt, ETL pipeline design, Azure Data Factory, Azure Synapse, Azure Blob Storage, Databricks, Git

Frontend & Product: React, REST API integration, FastAPI backend development

Analytics & Ways of Working: Power BI, Excel (Advanced), Agile / SAFe, cross functional stakeholder coordination, Jira, Confluence

PERSONAL PROJECTS

Agent Reliability Layer

- Designed a Semantic Policy Engine and an Agent State Bridge that checkpoint AI agent actions for human in the loop approval before they apply, with optimistic concurrency guards to catch conflicting concurrent agent actions.
- Refined the architecture through multiple rounds of adversarial review, narrowing the initial scope to Salesforce CRM record updates approved through a Slack based human in the loop workflow.

SupplyIQ

- Built a 16 model dbt medallion architecture on DuckDB simulating SAP MM, SD and WM operations, orchestrated end to end with Airflow and served through a FastAPI backend covering 14 endpoints.
- Built an 8 page React dashboard and added a retrieval augmented LangGraph advisor, powered by LLaMA 3.3 through Groq, that answers questions grounded in the live gold layer data.

PR Review Bot

- Built a GitHub App that reviews every pull request using three parallel LangGraph agents covering code quality, security and performance.
- Implemented per repository review memory with Supabase pgvector retrieval and multi turn chat through @mention commands, running at zero monthly cost on free tier infrastructure.

PROFESSIONAL EXPERIENCE

Mutares SE & Co. KGaA

Jul 2024 to Aug 2025

BI Developer / Analytics Engineer

Munich, Germany

- Unified SAP MM, SD, and FI data from 5+ acquired portfolio companies into a governed dbt medallion architecture on Databricks, enforcing 28 automated data quality tests and eliminating reconciliation disputes across 5+ reporting streams.
- Automated SAP extraction with parameterized Azure Data Factory pipelines and orchestrated Bronze to Gold processing in Apache Airflow, cutting transformation time by 50% and ensuring dashboards only ever served validated data.
- Engineered a reusable "Golden Standard" financial report in Python and Power BI, deployed across 5+ portfolio companies through Azure DevOps, standardizing group reporting and reducing manual effort by 40%.
- Automated report refresh and distribution with Git based CI/CD pipelines and Power Automate, reducing delivery failures by 80% (from 5 to 1 per month) and guaranteeing one consistent version for every stakeholder.
- Built an AI advisor with FastAPI and Llama 3.3 70B using hybrid RAG over Gold tables and SLA contracts in pgvector, determining SLA compensation for delayed orders from governed data without fine-tuning.

Mercedes-Benz Group AG

Oct 2023 to Mar 2024

*Business Analyst (Intern)**Stuttgart, Germany*

- Delivered Azure cloud cost allocation reporting for the Data Ecosystem team, giving platform owners spend visibility and enabling a 15% reduction in identified waste and 20% better allocation efficiency.
- Developed standardized Power BI dashboards across 5 departments, creating a single source of truth for controlling and quarterly planning KPIs.

Capgemini Technology Services

Sep 2019 to Aug 2022

*Senior Analyst / BI Developer**Bengaluru, India*

- Built and maintained ETL pipelines in SQL, Python, and SAP HANA feeding global financial KPI reporting, sustaining a 98.5% on time SLA across 3 years of production operations.
- Implemented change data capture with SAP SLT and Kafka streaming for sales orders, purchase orders, inventory, and sales and purchase accounts, cutting data latency by over 50% to 15 minutes.
- Designed SAP HANA Calculation Views and SDI flowgraphs to integrate and transform multiple source tables, improving query performance by over 60% for downstream reporting.
- Orchestrated SAP BW process chains from ingestion through a semantic reporting layer, automating recurring reporting processes and improving ETL efficiency by 20%.
- Delivered self service reporting through governed data views over SAP HANA, reducing business dependency on technical teams by 65%.

EDUCATION

M.Eng. Business Intelligence & Data Analytics | Hochschule Emden-Leer.

Sep 2022 to Aug 2025

Emden, Germany.

Overall grade 1.91 (gut) and thesis grade Thesis 1.30 (sehr gut) "Business Management Framework for Integrating Predictive Analytics into Financial Operations Using Power BI"

B.Eng. Mechanical Engineering | Visvesvaraya Technological University

Jun 2015 to Jun 2019

Belagavi, India

ADDITIONAL

Certification: Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate (valid to Sep 2027)

Languages: English (C1) · Kannada (native) · German (A2, actively improving)

Availability: Immediate, Germany